Biology (BSC-401)

Course Details

Credit Points: 3

Pre-Requisites

1. Basic knowledge of Physics, Chemistry and Mathematics

Syllabus Outline

Unit 1

- To convey that Biology is as important a scientific discipline as Mathematics, Physics and Chemistry.
- Fundamental differences between science and engineering drawn through comparisons (eye vs camera, bird flight vs aircraft).
- Biological observations of the 18th Century that led to major discoveries; examples from Brownian motion and thermodynamics (Robert Brown, Julius Mayer).

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Unit 2

Classification

- Hierarchy of life forms at the phenomenological level.
- Classification based on:
  (a) cellularity — unicellular or multicellular
  (b) ultrastructure — prokaryotes or eukaryotes
  (c) energy and carbon utilisation — autotrophs, heterotrophs, lithotrophs
  (d) ammonia excretion — amino telic, uricotelic, ureotelic
  (e) habitat — aquatic or terrestrial
  (f) molecular taxonomy — three major kingdoms of life. Model organisms: E. coli,
  S. cerevisiae, D. melanogaster, C. elegans, A. thaliana, M. musculus.

Unit 3

Genetics

- Mendel's laws, concept of segregation and independent assortment, alleles, gene mapping, gene interaction, epistasis.
- Meiosis and mitosis in the context of genetics.
- Concepts of recessiveness and dominance, mapping of phenotype to genes, single gene disorders in humans, complementation.

Unit 4

Biomolecules

- Molecules of life — monomeric units and polymeric structures.
- Sugars, starch and cellulose; amino acids and proteins; nucleotides and DNA/RNA; two-carbon units and lipids.

Unit 5

Enzymes

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- Monitoring enzyme-catalysed reactions.
- Mechanism of enzyme action (at least two examples).
- Enzyme classification.
- Enzyme kinetics and kinetic parameters.
- RNA catalysis.

Unit 6

Information Transfer

- DNA as genetic material.
- Hierarchy of DNA structure — single strand to double helix to nucleosomes.
- Concept of genetic code, universality and degeneracy.
- Definition of gene in terms of complementation and recombination.

Unit 7

Macromolecular Analysis

- Proteins — structure and function.
- Hierarchy in protein structure: primary, secondary, tertiary and quaternary.
- Proteins as enzymes, transporters, receptors and structural elements.

Unit 8

Metabolism

- Thermodynamics applied to biological systems.
- Exothermic/endothermic vs endergonic/exergonic reactions.
- Concept of k_{eq} and standard free energy.
- Spontaneity
- ATP as energy currency
- Breakdown of glucose to CO_2 + H_2O (Glycolysis and Krebs cycle) and synthesis of glucose from CO_2 and H_2O (Photosynthesis)
- Energy charge concept.

Unit 9

Microbiology

- Single-celled organisms — concept of species and strains
- Identification and classification of microorganisms
- Microscopy
- Ecological aspects of single-celled organisms
- Sterilization, media compositions, and growth kinetics.

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